C6.4 Transaction responses from the interconnect

The interconnect must provide a response for all transactions from an initiating master.

Table C6-2 shows the permitted response from the interconnect for a transaction issued on the AR channel.

Table C6-2 Permitted interconnect response for a transaction on the AR channel

Transaction from initiating master	Permitted response from the interconnect	
	IsShared, RRESP[3]	PassDirty, RRESP[2]
ReadNoSnoop	0	0
ReadOnce	0	0
	1	0
ReadClean	0	0
	1	0
ReadNotSharedDirty	0	0
	0	1
	1	0
ReadShared	0	0
	0	1
	1	0
	1	1
ReadUnique	0	0
	0	1
CleanUnique	0	0
MakeUnique	0	0
CleanShared	0	0
	1	0
CleanInvalid	0	0
MakeInvalid	0	0

The interconnect must determine the IsShared response for the following transactions from the initiating master:

- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared
- CleanShared.

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WasUnique：表示这个CacheLine在被Snoop之前是Unique状态的
IsShared：表示其他的Cache设备中也有这个CacheLine，所以自己的这个CacheLine的权限需要为Shared

After snooping all the required cached masters, the IsShared response to the initiating master for these transactions is determined as follows:

- If WasUnique was asserted for any snoop response received by the interconnect then:
  - If IsShared was asserted for that snoop response then, IsShared must be asserted in the transaction response to the initiating master. 如果一个Snoop请求得到的响应中IsShared有效，那么发送给master设备的IsShared也必须有效
  - If IsShared was deasserted for that snoop response then, this specification recommends that IsShared is deasserted in the transaction response to the initiating master. However, it is permitted to assert IsShared in the transaction response to the initiating master.

如果一个Snoop请求得到的响应中IsShared无效，那么发送给master中的IsSharead可以无效（推荐）

- If WasUnique was not asserted for any snoop response received by the interconnect then:
  - If any snoop responses had IsShared asserted then, IsShared must be asserted in the transaction response to the initiating master.
  - If all snoop responses received by the interconnect had IsShared and DataTransfer deasserted then, IsShared must be deasserted in the transaction response to the initiating master.
  - If all snoop responses received by the interconnect had IsShared deasserted and any snoop response had DataTransfer asserted then, this specification recommends that IsShared is deasserted in the transaction response to the initiating master. However, it is permitted to assert IsShared in the transaction response to the initiating master.

如果WasUnique无效：
- 如果有任意一个设备的Snoop响应将IsShared拉高那么给master设备的响应中的IsShared也要拉高
- 如果没有一个设备的Snoop响应将IsSharedlagao那么给master设备的响应中的IsShared需要拉低
- 如果Interconnect接收到的所有的Snoop响应中的IsShared无效，但是有任意一个Cache设备的DataTransfer有效，那么这个文档推荐你将返回给master设备的IsShared给拉低（也允许拉高）

The interconnect must determine the PassDirty response to the initiating master for the following transactions:

- ReadNotSharedDirty
- ReadShared
- ReadUnique.

After snooping all the required cached masters, the PassDirty response to the initiating master for these transactions is determined as follows:

- If PassDirty was asserted for any snoop response received by the interconnect, and the interconnect has not generated a write transaction to update main memory, then PassDirty must be asserted in the transaction response to the initiating master. 如果Snoop响应中存在PassDirty被拉高，并且Interconnect没有发起Write请求用于更新main memory，那么发送给master设备的响应中的PassDirty也要拉高

Note

Only transactions initiated on the AR channel have additional response bits returned with the transaction response to the initiating master on the R channel. 只有AR通道对应的响应通道R拥有这些响应信号

Write transactions do not have additional response bits, and the response from the transaction that passes through the interconnect can be returned directly to the initiating master. The write transactions are:

- WriteNoSnoop
- WriteUnique
- WriteLineUnique
- WriteBack
- WriteClean.

An Evict transaction does not propagate downstream and the interconnect is required to generate an OKAY, **BRESP[1:0]** = `0b00` write response.

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